10 — SEION-KGR v26 MAX: Engineering Closure of a Certified, Ablatable Knowledge-Graph Reasoner

A Gate 12 Closeout Report

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This is a submission-style writeup of what was built and executed, not a claim of state-of-the-art results. Read §10 (Limitations) before any other section if short on time.

Abstract

We report the engineering closure of SEION-KGR v26 MAX, a knowledge-graph link-prediction system combining reciprocal KGE base experts (DistMult/ComplEx/CP/TuckER), a budgeted-BFS query-conditioned path reasoner with both a fixed-random and a newly implemented learned edge selector, a CP-ternary "seionic" scalar branch, an orthogonal projection layer with six adaptive rank-allocation policies, a controlled structural-kernel residual branch (the real E_8 Lie-algebra structure-constant tensor, plus four matched controls: zero, random-scale-matched, permuted-index, sign-shuffled), and a state-to-score-to-ranking certification chain. All components are unit- and integration-tested (120 tests), CI-verified on real GitHub Actions infrastructure, and smoke- to screening-tested on real FB15K-237/WN18RR data using a real RTX PRO 5000 Blackwell GPU. We report a bounded, explicitly non-confirmatory 2-seed screening comparison of a base expert alone versus the same expert with the path and seionic branches enabled, two real negative-control integration runs (one showing the expected 4.1× MRR collapse under randomized training labels; one showing a null leakage effect that we investigate rather than accept, tracing it to an under-trained near-zero-init residual gate), and we report honestly that no Gate 12 confirmatory statistical campaign, no E_8-vs-controls comparison, and no full 13-point causal ablation ladder was executed at benchmark scale in this session. We do not claim state-of-the-art performance.

Contents

1 Introduction

Knowledge-graph embedding (KGE) systems predict missing edges in a knowledge graph. The SEION-KGR line of work (contract: `docs/SEION_KGR_MATHEMATICAL_CONTRACT.md`) proposes combining standard KGE experts with a ternary "seionic" algebraic law, projection-based compression with certified error propagation, and an optional structural-kernel residual inspired by the E_8 Lie algebra's structure constants. Prior work on this repository (canonical commit `6af3c35271ae2ffab41ecba2aad098d1988fdc0c`) delivered an implementation-complete, smoke-verified system across nine build phases plus checkpointing. This report documents the subsequent `gate12-closeout` campaign: closing the remaining engineering gaps that prior work explicitly left open, and executing as much real benchmark evidence as a single session's compute budget allowed.

2 Related problem setting and scope

We do not attempt a literature-comparable benchmark run in this campaign (§10). The base experts implemented (DistMult, ComplEx, CP/canonical-decomposition, TuckER) are standard; readers should consult the original papers for those architectures. The path reasoner is a budgeted-BFS message passer in the NBFNet/A*Net family, explicitly *not* claimed to faithfully reproduce either (the repository's own naming convention forbids borrowing those names for an approximation, per its engineering mandate).

3 Mathematical formulation

See `02_mathematical_foundations_and_claim_status.pdf` for the complete, precisely labeled statement. In summary: a typed CP ternary law $\mu(x, a, q) = O[(Ax) \odot (Ba) \odot (Cq)]$, proved equal to its dense tensor contraction; an orthogonal projector $P = QQ^*$ via Stiefel-manifold QR retraction; the exact local subset-expansion identity for typed multilinear trees (inherited, PROVED, with a new conformance test against the concrete implementation); the $(k-1)$ projected-root error coefficient (inherited, PROVED_UNDER_ASSUMPTIONS); and this campaign's new state→score→ranking certification chain, which certifies only the case where nothing is approximated in the score path.

4 SEION-KGR architecture

$$s(h, r, t) = s_{\text{base}}(h, r, t) + \gamma_r s_{\text{path}}(h, r, t) + \eta_r s_{\text{seion}}(h, r, t) + s_{\text{kernel}}(h, r, t),$$

every experimental term gated near zero at initialization. Reciprocal relations collapse head-ranking into tail-ranking of the inverse relation — one evaluator code path, not two. Full module map in `03_system_architecture_and_implementation.pdf`.

4.1 Learned path selector (new this campaign)

`LearnedPathSelector` scores each candidate outgoing edge from source state, edge relation, query relation, destination base embedding, depth, and accumulated path priority; the top-k kept edges are weighted by a differentiable σ(score) gate. Verified: exact budget compliance, determinism under fixed inputs, an explicit tie policy (lowest original index wins under an exact tie), permutation invariance of neighbor ordering, gradients reaching the selector's parameters, and — the property that matters most — gold-path preservation after 200 steps of supervised warmup on a synthetic fixture where only one of six candidate edges is correct.

4.2 Structural kernel residual (new this campaign)

Five variants of $\mu_{\text{total}} = \mu_{\text{learned}} + \sigma(\epsilon_r)\, W(\mu_{\text{kernel}}(U_x x, U_a a, U_q q))$ share an identical architecture and trainable-parameter count: the real E_8 structure-constant tensor (`E8_Exact_v18_2/f_E8.npy`, 248^3, Killing rank 248, Jacobi residual $\sim 2\times10^{-13}$ per its own provenance record) and four matched controls. Every kernel gets a SHA-256 + Frobenius-norm + source provenance record.

5 Error and ranking certification

See `07_certification_error_bounds_and_rank_stability.pdf`. We certify exactly the case where no approximation occurs in the score path; we *correctly refuse* to certify the moment a rank-reducing projector or the path reasoner's nonlinear envelope is present, verified with a margin of 2000 that still fails to certify — proving the refusal is assumption-driven, not margin-driven.

6 Experimental protocol

Preregistered in `campaigns/gate12/preregistration.md` before any Phase C result was generated, including an explicit compute-budget declaration stating that the full 3-seed screening/5-seed confirmatory protocol across 13 ablation points would not be executed this session. What was executed instead (Tier 0, defined and frozen before results were seen): a 1-epoch, 4-way base-expert screening pass on FB15K-237 (selecting TuckER), followed by a 2-seed comparison of TuckER alone (A0, full dataset) versus TuckER+path+seion (A3, 5,000-triple subsample — a compute-budget deviation logged *before* this A3 run, after a live timing probe showed the full-scale configuration did not complete a single epoch within 8 minutes).

7 Results

See `04_gate12_confirmatory_benchmark_report.pdf` for the numbers in full, and `11_supplement.pdf` for every seed-level value. Headline (screening-tier, 2 seeds, non-confirmatory, A0/A3 confounded by the training-subset-size difference — see §10): base-expert screening selected TuckER on FB15K-237 valid MRR (0.2534 vs. ComplEx 0.2210, DistMult 0.2195, CP 0.2092, 1 epoch, capped eval).

8 Ablations and controls

2 of the preregistered 13 causal-ladder points executed at dataset scale (A0, A3); the remaining 11, plus the 5-point E8/control block, are implemented and unit-tested but not run at dataset scale this campaign. Full breakdown: `05_ablations_and_causal_controls.pdf`.

9 Efficiency and compression

Base-expert-only training on full FB15K-237 (dim 64, GPU): ∼80s for 5 epochs including evaluation. The path-reasoner-enabled configuration does not scale to full-dataset batches within this session's compute budget (§10) — a real, measured, architectural finding (per-sample Python-loop BFS, GPU actively *hurts* due to kernel-launch overhead on tiny per-sample operations, consistent with this repository's own prior finding of 3.2–3.5× GPU slowdown on small operations).

10 Limitations and negative results

Full detail: `09_negative_results_limitations_and_open_problems.pdf`. Headline items: (1) the path-reasoner residual gate did not measurably open in a real 40-epoch training run, confounding the negative-control leakage test and, more importantly, raising an open question about whether the near-zero-init router design activates within realistic training budgets at all; (2) the path reasoner does not scale to full benchmark-dataset batches within a session's compute budget; (3) certification coverage is honestly 0% for any model using the path reasoner; (4) no E_8-causal-attribution evidence was collected; (5) no confirmatory statistical campaign was run.

11 Conclusion

This campaign closes the engineering gaps the canonical commit explicitly left open (learned selector, E_8 residual with matched controls, certification chain, CI, several reproducibility correctness bugs caught and fixed by real execution rather than inspection) and executes a bounded, honestly-labeled first look at benchmark-scale behavior. It does not, and should not be read to, establish that any of SEION-KGR's experimental branches improve knowledge-graph link prediction. That question is open, and §10 states plainly what would be required to answer it.

A Appendix: reproduction

`08_reproducibility_software_and_artifacts.pdf` and `11_supplement.pdf` carry every command and seed-level value needed to regenerate every number in this document.